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A Survey of LLMs for Autonomous Driving

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Abstract

The rapidly evolving fields of autonomous driving (AD) and Large Language Models (LLMs) represent two of the most transformative technological advancements of our time. While autonomous vehicles promise to revolutionize transportation, they face formidable challenges related to complex reasoning, human interaction, and safety in highly dynamic environments. Concurrently, LLMs have demonstrated unprecedented capabilities in understanding, generating, and reasoning with human language, offering a powerful new paradigm for intelligent systems. This survey establishes the motivation for exploring the convergence of these two powerful technologies, emphasizing the potential for LLMs to address some of the most intricate challenges in AD. We provide a comprehensive overview of LLM fundamentals and the core components of autonomous driving systems, bridging these domains to illustrate their inherent complementarity. Furthermore, we systematically analyze various methodologies and architectural patterns for integrating LLMs into AD, categorizing their key applications across perception, prediction, planning, control, and human-vehicle interaction. Finally, we critically examine the current challenges, limitations, and ethical considerations impeding widespread deployment, offering promising future research directions to advance the field towards safer, more intelligent, and human-centric autonomous systems.

Keywords

autonomous driving, autonomous systems, computing and processing, human-centric autonomous systems, large language models

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Abstract—The rapidly evolving fields of autonomous driving (AD) and Large Language Models (LLMs) represent two of the most transformative technological advancements of our time. While autonomous vehicles promise to revolutionize transportation, they face formidable challenges related to complex reasoning, human interaction, and safety in highly dynamic environments. Concurrently, LLMs have demonstrated unprecedented capabilities in understanding, generating, and reasoning with human language, offering a powerful new paradigm for intelligent systems. This survey establishes the motivation for exploring the convergence of these two powerful technologies, emphasizing the potential for LLMs to address some of the most intricate challenges in AD. We provide a comprehensive overview of LLM fundamentals and the core components of autonomous driving systems, bridging these domains to illustrate their inherent complementarity. Furthermore, we systematically analyze various methodologies and architectural patterns for integrating LLMs into AD, categorizing their key applications across perception, prediction, planning, control, and human-vehicle interaction. Finally, we critically examine the current challenges, limitations, and ethical considerations impeding widespread deployment, offering promising future research directions to advance the field towards safer, more intelligent, and human-centric autonomous systems.

I. INTRODUCTION

Autonomous driving (AD) systems are poised to revolutionize transportation, offering profound benefits such as enhanced safety, reduced traffic congestion, and improved mobility. Despite significant progress, achieving fully autonomous navigation in complex, real-world scenarios remains a formidable challenge. Autonomous vehicles must accurately perceive their surroundings, predict the behavior of other road users, make intelligent and safe planning decisions, and execute precise control actions. These tasks demand sophisticated reasoning capabilities, robust perception under uncertainty, and seamless interaction with human drivers and pedestrians [1].

Concurrently, Large Language Models (LLMs) have emerged as a groundbreaking force in artificial intelligence, demonstrating exceptional performance across a wide spectrum of Natural Language Processing (NLP) tasks. Fueled by massive datasets and transformer architectures, LLMs exhibit remarkable abilities in language understanding, generation, and complex reasoning [2], [3]. Their capacity to process and generate human-like text, coupled with emergent reasoning skills, presents a compelling opportunity to address the intricate cognitive demands of autonomous driving.

This survey aims to explore the convergence of LLMs and autonomous driving, specifically investigating how the advanced reasoning, understanding, and generative capabilities of LLMs can be leveraged to enhance AD systems. We establish the motivation for this integration, highlighting the potential

for LLMs to contribute to high-level decision-making, natural human-vehicle interaction, and robust scene understanding, areas where traditional, modular AD approaches often face limitations.

The objective of this paper is to provide a comprehensive overview of the state-of-the-art in integrating LLMs into autonomous driving. We delve into various integration methodologies, architectural paradigms, and practical applications within the AD stack. We also critically examine the current challenges, limitations, and ethical considerations inherent in deploying LLMs in safety-critical domains, and outline promising future research directions. The structure of this survey is as follows: Section II provides foundational knowledge on both LLMs and AD. Section III details the technical approaches for integrating LLMs into AD systems. Section IV categorizes and describes key applications. Section V discusses challenges, limitations, and future directions. Finally, Section VI concludes the survey.

II. BACKGROUND ON LLMs AND AUTONOMOUS DRIVING

To understand the synergistic potential of LLMs and autonomous driving, it is essential to establish a foundational understanding of each domain. This section provides a concise overview of Large Language Models, their core capabilities, and the fundamental architecture and challenges of autonomous driving systems.

A. Large Language Models: Architectures, Capabilities, and Evolution

Large Language Models (LLMs) are a class of deep neural networks, primarily based on the Transformer architecture, that have achieved unprecedented success in a wide range of natural language tasks [2]. Trained on vast quantities of text data, these models learn intricate patterns of language, enabling them to understand context, generate coherent text, and even perform complex reasoning.

The Transformer architecture, with its self-attention mechanisms, allows LLMs to process input sequences in parallel and capture long-range dependencies, overcoming limitations of previous recurrent neural network (RNN) based models. The pre-training paradigm, typically involving masked language modeling or next-token prediction, endows LLMs with a broad understanding of world knowledge and linguistic structures [2]. Following pre-training, LLMs are often fine-tuned on specific tasks or aligned with human preferences through techniques like Reinforcement Learning from Human Feedback (RLHF), leading to models like ChatGPT and GPT-4 that exhibit impressive conversational abilities [4].

LLMs demonstrate diverse capabilities:

- **Language Understanding and Generation:** LLMs can parse complex natural language queries, extract information, and generate fluent, coherent, and contextually relevant responses. This ability is crucial for tasks like abstractive summarization [5]–[7] and multilingual machine translation [8]–[10].
- **Reasoning:** LLMs exhibit emergent reasoning capabilities, particularly with techniques like Chain-of-Thought (CoT) prompting, enabling them to break down complex problems into intermediate steps [11], [12]. This reasoning can be further enhanced through self-improvement mechanisms [3] or alignment fine-tuning [13]. Their ability to reason over structured data is also being explored [14]. Furthermore, recent work investigates LLMs as models for analogical reasoning, mirroring human cognitive processes [15].
- **Evaluation and Benchmarking:** The evaluation of LLMs is an active area of research, with benchmarks like MEGA [2] assessing multilingual performance and frameworks like G-Eval [16] proposing LLM-based evaluation that correlates well with human judgments [17]. Beyond traditional NLP tasks, LLMs are being evaluated for their capacity in specialized tasks like sentiment analysis [18] and as re-ranking agents for search [19], [20].
- **Fine-tuning and Efficiency:** Adapting LLMs to new tasks or domains efficiently is critical. Parameter-Efficient Fine-Tuning (PEFT) methods, such as those integrated in LLM-Adapters [21], allow for substantial performance gains with minimal trainable parameters. Strategies include partial fine-tuning [22], federated sketching LoRA [23], and fine-tuning with very large dropout for robustness [24]. The efficiency of LLM inference on various hardware, including CPUs, is also a significant area of development [25], [26].
- **Multimodal Capabilities:** A rapidly growing area involves multimodal LLMs (MLLMs), which integrate language with other modalities like vision [27]–[29]. These models can perform tasks like Visual Question Answering (VQA) [30]–[32] and are foundational for processing diverse sensor inputs in AD.

Despite their impressive abilities, LLMs also face limitations, such as potential for deception [33], lack of full understanding of character composition [34], and vulnerabilities to attacks like jailbreaking [35]. However, ongoing research into self-cognition [36] and enhancing human-like responses [37], [38] continues to push their boundaries. Furthermore, LLMs’ versatility extends to diverse analytical domains, such as time series forecasting [39] and even modeling economic growth for small and medium-sized enterprises (SMEs) [40]–[42], showcasing their general utility as AI agents [43]. The internal representations of LLMs are also being studied, revealing parallels with human cognitive biases, such as logarithmic number perception [44].

B. Autonomous Driving Systems: Components, Challenges, and Current Approaches

Autonomous driving systems typically comprise a modular pipeline, each component addressing a critical aspect of vehicle operation: perception, prediction, planning, and control.

- **Perception:** This module uses various sensors to understand the vehicle’s surroundings. Common sensor modalities include cameras, LiDAR (Light Detection and Ranging), and radar [45], [46]. Perception tasks involve 3D object detection [47], semantic and instance segmentation, and scene understanding [48]–[50]. Robustness to environmental factors like soiling is also a challenge [51]. Cooperative perception, leveraging Vehicle-to-Vehicle (V2V) communication, can enhance perception, even under lossy communication conditions [52].
- **Prediction:** The prediction module forecasts the future trajectories and intentions of other dynamic agents (vehicles, pedestrians, cyclists) in the environment. This is crucial for safe planning, considering complex social interactions and traffic rules [53]–[56]. Methods often involve deep learning to model dynamic social interactions and conform to semantic maps [57].
- **Planning:** This module generates a safe and efficient path for the autonomous vehicle, considering perceived objects, predicted behaviors, traffic rules, and high-level goals. Planning involves both global route planning and local trajectory generation. Deep Reinforcement Learning (DRL) frameworks are increasingly used for decision-making in complex driving scenarios [58], [59]. Multi-agent RL is particularly relevant for handling interactions in mixed traffic [60], [61].
- **Control:** The control module executes the planned trajectory by adjusting steering, acceleration, and braking. Control strategies, such as PID (Proportional-Integral-Derivative) and pure pursuit algorithms, are fundamental to achieving accurate and stable vehicle dynamics [1], [62], [63]. Personalization in control systems can also provide a more comfortable driving experience [64].

Supporting infrastructure includes the Internet of Autonomous Vehicles (IoAV) architecture for ubiquitous connectivity [65] and real-time monitoring systems [66]. The development of AD systems relies heavily on extensive datasets [45] and simulation environments for testing and validation [67]–[70]. Ensuring safety, interpretability, and regulatory compliance is paramount [71]–[73]. General AI concepts, like modular mechanistic networks, contribute to understanding deep learning models in NLP, which can inform AD development [74].

C. Bridging LLMs and Autonomous Driving

The convergence of LLMs and autonomous driving stems from a recognition that LLMs’ advanced capabilities can address critical gaps in traditional AD paradigms. While classical AD modules excel at low-level tasks, they often struggle with nuanced human-like reasoning, ambiguous situations, and intuitive human interaction. LLMs, with their ability to process vast amounts of unstructured data and infer complex semantic relationships, are well-positioned to augment AD systems

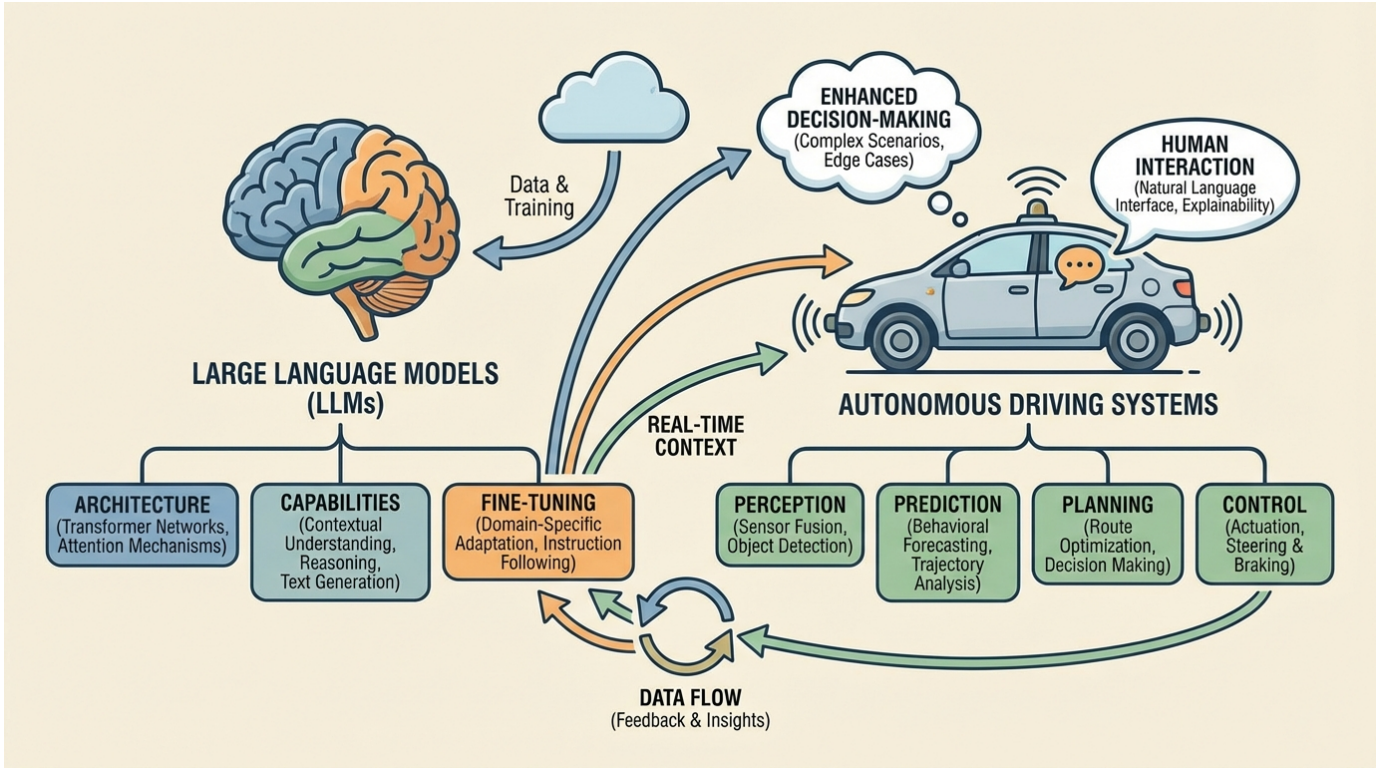


Fig. 1. Overview of the synergistic relationship between Large Language Models (LLMs) and Autonomous Driving Systems, highlighting key architectural components, capabilities, and interaction pathways for enhanced decision-making and human interaction.

in areas such as high-level decision-making, contextual understanding, and natural language interfaces. The potential to translate human intent, explain complex decisions, and even simulate human-like behaviors makes LLMs a promising avenue for developing more robust, flexible, and human-centric autonomous vehicles. The fundamental challenge lies in effectively grounding the LLMs' linguistic and reasoning capabilities in the dynamic, physical world of autonomous driving.

III. METHODOLOGIES AND ARCHITECTURES FOR INTEGRATING LLMs IN AD

Integrating Large Language Models into autonomous driving systems requires novel methodologies and architectural adaptations to bridge the gap between high-level linguistic reasoning and low-level physical control. This section explores various paradigms and techniques for incorporating LLMs into the AD stack.

A. LLMs as High-Level Planners and Reasoners

One prominent approach is to leverage LLMs for high-level planning and reasoning, essentially positioning them as the "brain" or a cognitive layer for autonomous vehicles.

- **Symbolic Planning and Decision-Making:** LLMs can be prompted to generate symbolic plans or sequences of actions based on textual descriptions of scenarios and goals. Frameworks like DualAD [75] propose a dual-layer planning architecture where an LLM in the upper layer intervenes in

rule-based motion planning, mimicking human reasoning in critical situations. The integration of LLMs in hierarchical planning, where they assist in generating high-level tasks or decomposing problems, is also a promising area [76]. Evaluations specifically target LLMs' capacity for spatial-aware decision-making and adherence to traffic rules, crucial for autonomous driving [77]. The concept of Reasoning via Planning (RAP) repurposes LLMs as both world models and reasoning agents, using algorithms like Monte Carlo Tree Search for strategic exploration of reasoning paths, enhancing performance in complex tasks like plan generation [11].

- **Neuro-Symbolic Approaches:** Combining the strengths of connectionist models (like LLMs) and symbolic reasoning systems offers a robust path forward. Approaches such as DRL with Symbolic Logic (DRLSL) integrate DRL with first-order logic to enable safer learning policies in real-time AD interactions [78]. LLMs can contribute to the symbolic component by generating logical rules, semantic descriptions, or abstract plans that guide the underlying DRL or control systems.

B. Natural Language Interfaces and Human-Vehicle Interaction

LLMs are particularly well-suited for enhancing human-vehicle interaction (HVI) by enabling natural language (NL) communication.

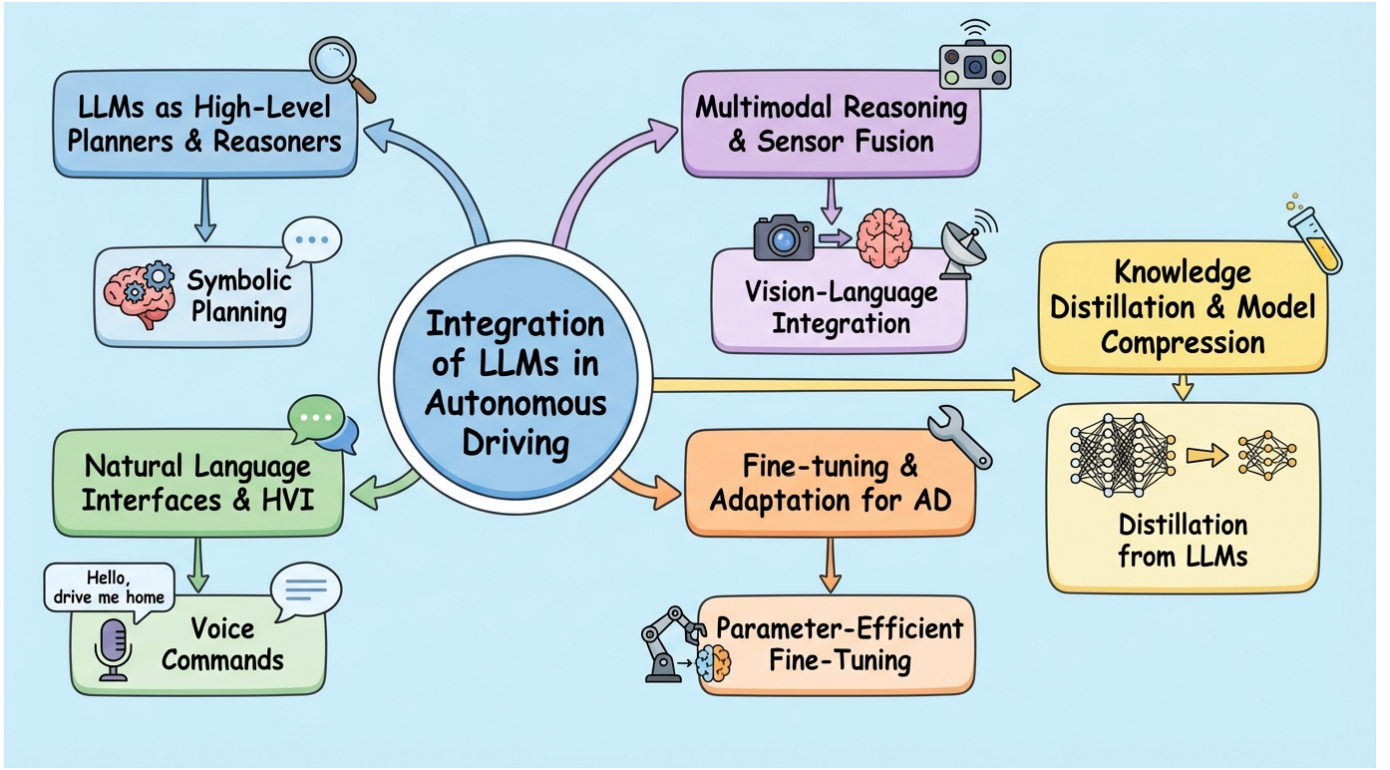


Fig. 2. Overview of methodologies and architectural paradigms for integrating Large Language Models (LLMs) into Autonomous Driving (AD) systems. This figure illustrates key areas such as high-level planning, human-vehicle interaction, multimodal reasoning, fine-tuning strategies, and model compression techniques.

- **Voice Commands and Instructions:** LLMs can interpret complex voice instructions from occupants, allowing for intuitive control and task specification. Research has explored incorporating natural language voice instructions into model-based reinforcement learning for self-driving cars, significantly boosting learning speed [79]. This extends the utility of LLMs beyond traditional text-based NLP tasks to real-time interaction, similar to leveraging pre-trained language models for natural language interaction in self-tracking applications [80].
- **Generating Explanations:** For safety-critical systems like autonomous vehicles, trust and transparency are paramount. LLMs can generate human-centric causal explanations for the vehicle’s motion planning decisions in natural language [81]. These explanations can be holistic, aligned with perception-prediction-planning outputs, as seen in Hint-AD [82]. The ACE paradigm (Action and Control via Explanations) proposes using LLMs to produce human-interpretable “semantic frames” to align multimodal AI assistants and support human performance [83], a concept directly applicable to AD explainability. This capability is critical for building trustworthy and interpretable LLMs [84].
- **Personalized Interaction:** LLMs can adapt to individual driving preferences and communication styles, offering a more personalized and comfortable driving experience [64]. Data collection on human driving behavior, including multimodal databases with facial video records, is crucial for

understanding and characterizing drivers’ affective states and styles for such personalized systems [85]. Validating driver models through human factors approaches is essential for interaction-aware automated vehicles [86].

C. Multimodal Reasoning and Sensor Fusion

The real world is inherently multimodal, and autonomous vehicles rely on diverse sensor inputs. LLMs, particularly multimodal variants (MLLMs), are increasingly being adapted to process and reason across these modalities.

- **Vision-Language Integration:** MLLMs can combine visual information from cameras with linguistic queries to perform tasks like Visual Question Answering (VQA) in driving scenes [87]. OpenEMMA [88] presents an open-source multimodal model for end-to-end autonomous driving, leveraging MLLMs and Chain-of-Thought reasoning. Integrating vision into generative pre-trained language models for multimodal summarization demonstrates how visual information can guide text generation [7].
- **Sensor Fusion and Holistic Understanding:** DriveAgent [89] proposes a multi-agent AD framework that combines LLM reasoning with multimodal sensor fusion (camera, LiDAR, GPS, IMU) for enhanced situational understanding and decision-making. The LLM coordinates specialized perception and reasoning agents to deliver cohesive, interpretable insights. While not LLM-specific, broader research in multimodal sentiment analysis explores methods for fus-

ing linguistic, visual, and acoustic features, tackling issues like modality robustness [90], contrastive learning [91], [92], language-guided representation learning [93], graph neural networks [94], and modal-temporal attention graphs [95]. These methods offer blueprints for sophisticated sensor fusion in AD. Efforts in developing scalable primitives for generalized sensor fusion in autonomous vehicles are also crucial [46].

- **Handling Missing Modalities:** In real-world AD scenarios, sensors can fail or be occluded. Techniques for handling missing modalities, such as imagination networks for emotion recognition [96], could inspire solutions for robust perception in AD when sensor data is incomplete. Studies revisiting the need for visual context in multimodal machine translation highlight the challenges and interpretability issues in multimodal integration [97]. Research on inertial and mechanomyography-based sensor fusion for wearable facial activity recognition also showcases advanced sensor fusion beyond traditional AD contexts [98].

D. Fine-tuning and Adaptation Strategies for AD

Adapting general-purpose LLMs for specific AD tasks efficiently and effectively is critical.

- **Parameter-Efficient Fine-Tuning (PEFT):** Methods like LLM-Adapters [21] enable fine-tuning LLMs on AD-specific datasets with a minimal number of trainable parameters, making adaptation more cost-effective. Federated Sketching LoRA [23] provides a flexible framework for heterogeneous collaborative fine-tuning of LLMs across resource-constrained clients, which could be relevant for distributed AD development.
- **Data-Efficient Fine-Tuning:** Given the high cost of AD data annotation, data-efficient fine-tuning strategies are essential. Techniques like compositional fine-tuning [99], where LLMs are fine-tuned on a curriculum of component tasks, can improve performance with less data. Partial fine-tuning [22] and human-inspired learning strategies for fine-tuning efficiency [100] are also explored in general NLP, applicable to AD.
- **Structure-Aware Fine-Tuning:** For tasks involving structured inputs like Abstract Meaning Representations (AMRs), structure-aware fine-tuning (SAFT) can inject graph topology into LLMs without architectural changes, improving performance [101]. This approach could be adapted for integrating structured AD knowledge (e.g., traffic rules, road network graphs) into LLMs.
- **Continual Learning:** Autonomous vehicles operate in dynamic environments where new situations constantly arise. Continual learning, which allows models to adapt to new tasks without catastrophically forgetting previous knowledge, is crucial. Densely Distilling Cumulative Knowledge (DKD) addresses this by distilling knowledge of all previous tasks, outperforming state-of-the-art baselines [102]. This can ensure that LLM-driven AD agents can continuously learn from new experiences. The ability of legged robots to

fine-tune locomotion policies in the real world [103] serves as an inspiration for real-world continuous adaptation.

- **Privacy-Preserving Fine-tuning:** Addressing privacy concerns in AD, particularly when training on sensitive data, is vital. Differentially private fine-tuning algorithms are being developed for LLMs to achieve state-of-the-art privacy-utility tradeoffs on standard NLP tasks [104].

E. Knowledge Distillation and Model Compression

Large Language Models are computationally intensive, making their direct deployment on resource-constrained autonomous vehicles challenging. Knowledge distillation (KD) offers a solution by transferring knowledge from a large, complex teacher model (e.g., a powerful LLM) to a smaller, more efficient student model.

- **Distillation from LLMs:** The concept of distilling knowledge from LLMs has been explored in various domains, such as household energy modeling [105], indicating its potential for AD. This involves extracting nuanced insights or reasoning capabilities from a large LLM and compressing them into a specialized, smaller model suitable for real-time AD tasks. Agentic scientific corpus distillation frameworks are also proposed for biomedical LLM training, emphasizing knowledge-driven extraction and self-evaluation [106].
- **General KD Techniques:** A wide array of KD techniques exists, including response-based, feature-based, and relation-based methods [107]. These can involve learning feature maps from the teacher network [108], revisiting KD via label smoothing regularization [109], or using multi-teacher distillation to aggregate knowledge from multiple sources [110]. Differentiable feature aggregation search methods also offer efficient ways to find optimal aggregations for knowledge transfer [111]. KD is also applied in multi-task learning to address imbalance problems across tasks [112]. These general KD methods can be adapted to compress LLM-derived knowledge into specialized perception, prediction, or planning modules for AD.

IV. KEY APPLICATIONS OF LLMs IN AUTONOMOUS DRIVING

The unique capabilities of Large Language Models position them to augment various aspects of autonomous driving, ranging from enhanced scene understanding to more intuitive human-vehicle interaction and sophisticated planning. This section systematically categorizes and details the diverse applications where LLMs are making significant contributions.

A. Advanced Scene Understanding and Semantic Reasoning

LLMs can elevate the perception and understanding capabilities of autonomous vehicles beyond mere object detection to a rich, semantic comprehension of driving scenarios.

- **Contextual and Causal Reasoning:** LLMs can process raw sensor data combined with textual prompts to understand complex environmental contexts, infer intentions, and identify causal relationships between events. For instance, they can answer questions about why an event occurred or

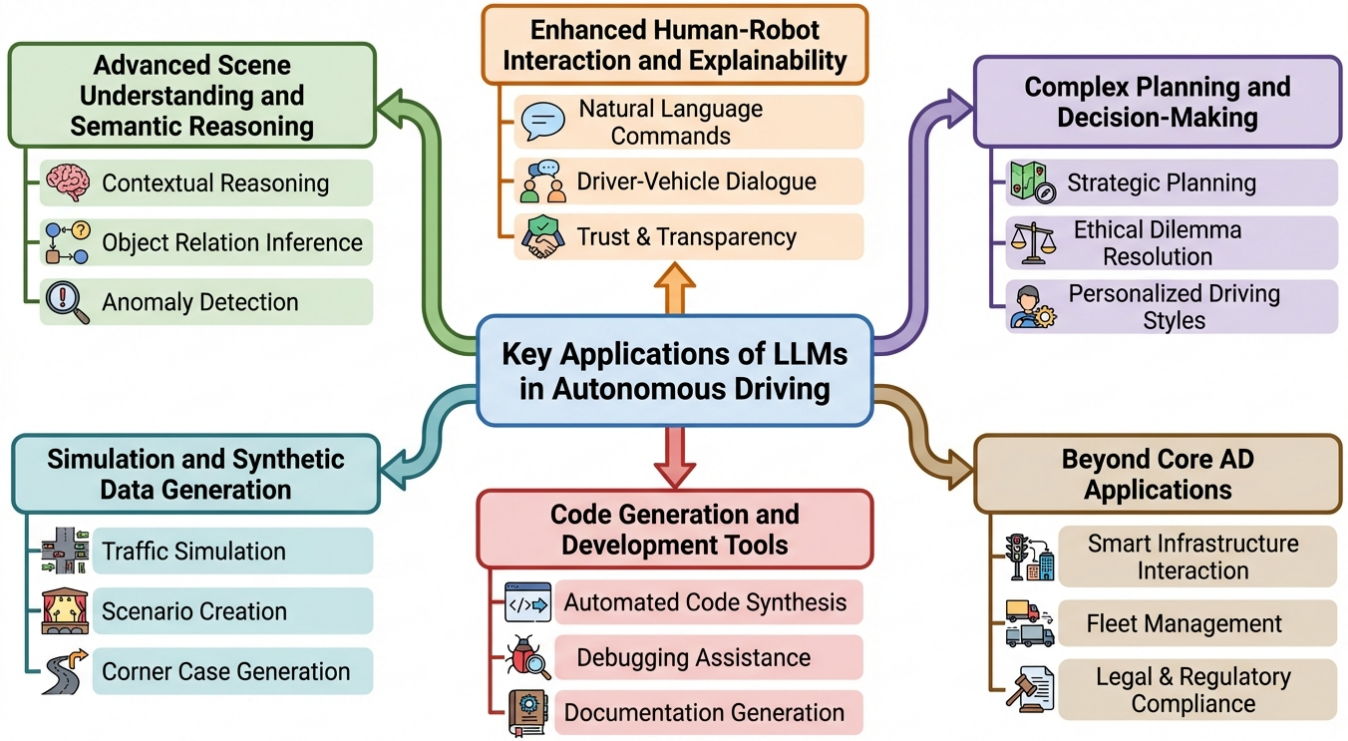


Fig. 3. An overview of key applications of Large Language Models (LLMs) in autonomous driving. The figure categorizes LLM applications into advanced scene understanding, human-robot interaction, complex planning and decision-making, simulation and synthetic data generation, code generation and development tools, and broader supportive applications.

what might happen next, moving beyond reactive systems to proactive reasoning. DriveAgent [89] demonstrates LLM-driven analytical processes structured across specialized agents to explain contextual changes and their underlying mechanisms. Similarly, research into causal reasoning benchmarks for LLMs highlights their potential to move beyond retrieval of domain knowledge to true interventional or counterfactual reasoning. This capability aligns with developing context-aware AI interventions that support effective reasoning [113].

- **Vision-Language Question Answering:** By integrating vision and language, LLMs can provide detailed textual descriptions or answer specific questions about driving scenes. Multi-Frame, Lightweight & Efficient Vision-Language Models (EM-VLM4AD) perform Visual Question Answering for autonomous driving, extracting relevant information from traffic views related to prompts [87]. This capability is extended by models that can holistically align interpretability in end-to-end autonomous driving, generating language explanations grounded in perception-prediction-planning outputs [82].

B. Enhanced Human-Robot Interaction and Explainability

LLMs are transforming how humans interact with autonomous vehicles, enabling more natural communication and fostering trust through transparency.

- **Natural Language Commands and Querying:** Instead of rigid command interfaces, users can issue instructions to autonomous vehicles using natural language, including voice commands [79], [80]. This allows for more intuitive control and task specification, mirroring the ease of interacting with traditional natural language database interfaces [114], [115].
- **Generating Explanations for AD Decisions:** Crucially, LLMs can explain the autonomous vehicle's actions and decisions in understandable human language [81]. This interpretability is vital for user trust and acceptance in safety-critical applications. The ACE paradigm, focusing on Action and Control via Explanations, proposes LLMs to provide human-centered explainability for multimodal AI assistants, directly applicable to AD [83].
- **Personalized Driving Experience:** LLMs can learn and adapt to individual driver preferences, creating a personalized and comfortable driving style, moving beyond a generic "robocar" experience [64]. Research on human-vehicle interaction (HVI) in real and virtual reality-based scenarios helps assess behavioral gaps and validate such personalized systems [116]. Multi-modal facial video databases are also being developed to characterize drivers' affective states and driving styles for enhanced HVI research [85].

C. Complex Planning and Decision-Making

LLMs can contribute to high-level strategic planning and complex decision-making, particularly in ambiguous or multi-

agent scenarios.

- **Strategic Planning and Reasoning:** LLMs can be used to generate robust plans and strategies for autonomous vehicles in intricate urban environments, considering potential risks and long-term consequences. DualAD [75] exemplifies LLMs serving as the upper layer for reasoning in a dual-layer planning framework. Initial evaluations confirm LLMs' capabilities in spatial-aware decision-making and adhering to traffic rules [77]. A roadmap guides the integration of LLMs in hierarchical planning, acknowledging their nascent but promising capabilities [76].
- **Multi-Agent Coordination:** In environments with multiple autonomous vehicles and human-driven vehicles, LLMs can facilitate coordination and cooperative decision-making. Multi-agent reinforcement learning frameworks are developed for cooperative lane changing in mixed traffic [61] and for general multi-agent connected autonomous driving [60]. LLMs can potentially guide these agents by providing high-level communication and shared understanding of complex goals.

D. Simulation and Synthetic Data Generation

High-fidelity simulation and vast amounts of diverse training data are indispensable for developing and validating AD systems. LLMs offer new avenues for generating realistic scenarios and data.

- **Human-like Traffic Simulation:** LLMs can be leveraged to create more realistic and diverse human-like traffic behaviors within simulation environments. HDSim [117] introduces an LLM-based framework that combines cognitive theory with LLM assistance to produce scalable and realistic traffic scenarios for self-driving tests, significantly improving the detection of safety-critical failures. This represents a significant advancement over heuristic or narrow data-driven models.
- **Synthetic Scene Generation and Data Augmentation:** Generative models, including those related to LLMs, can synthesize diverse driving scenes, weather conditions, and object variations. FreeGen [70] proposes a feed-forward reconstruction-generation co-training framework for free-viewpoint driving scene synthesis, crucial for closed-loop simulation. Techniques similar to GANs for LiDAR sensor modeling and data augmentation [118] can be extended. The broader field of generative AI is exploring techniques for creating personalized images and videos [119]–[123], which can inspire generating diverse visual data for AD.
- **Closed-Loop Benchmarking:** LLMs can assist in defining and evaluating complex closed-loop benchmarks for end-to-end autonomous driving, ensuring that simulations accurately reflect real-world conditions [69]. This is crucial for verifying and validating AD systems, particularly at challenging locations like intersections [73]. Agent-based simulation, as seen in retail management contexts [124], [125] or patient flow analysis [126], provides a paradigm for designing LLM-driven agents for complex simulations.

E. Code Generation and Development Tools

LLMs are proving to be powerful assistants in software development, including for AD systems.

- **Automatic Code Generation:** LLMs can generate code from natural language descriptions, accelerating the development of AD modules. This includes Java code generation [127], [128], which can be extended to other programming languages relevant to AD. The evolving role of programming and LLMs in self-driving laboratories highlights their potential in automating scientific research and experimental design [129].
- **Developer Productivity and Tooling:** LLMs can assist developers in writing, debugging, and optimizing code for autonomous vehicles. Frameworks like ChatDev [130], which uses communicative agents for software development, demonstrate how LLMs can streamline the entire development process. Benchmarks for tool-augmented LLMs, such as API-Bank [131], are crucial for evaluating their effectiveness in real-world engineering tasks.

F. Beyond Core AD Applications

While the primary focus is on AD, LLMs' general capabilities extend to areas that indirectly support or inform AD development.

- **Information Retrieval and Synthesis:** LLMs' ability to synthesize information [43] can be used for accessing and integrating vast amounts of AD-related knowledge, such as traffic laws, safety regulations, and operational manuals. They can serve as effective reranking agents for search results [19], [20], assisting AD engineers in finding relevant research or data.
- **Multilingual Support:** Given the global nature of autonomous vehicle deployment, LLMs' multilingual capabilities [2], [8] are crucial for understanding and interacting with drivers and regulations across different linguistic contexts [10]. This also extends to challenges in natural language processing for various languages [132], [133].
- **General AI Enhancements:** The use of LLMs in sentiment analysis [18] or personalized recommendations [134] showcases their ability to understand human intent and preferences, which is transferable to building more empathetic and responsive AD systems. Furthermore, general improvements in NLP tasks like automated multiple-choice question generation [135] can support training and assessment of human operators and developers within the AD ecosystem. The development of cross-domain adaptive templates for ML models can also contribute to the flexibility of AD systems [136].

V. CHALLENGES, LIMITATIONS, AND FUTURE DIRECTIONS

Despite the promising applications, integrating Large Language Models into autonomous driving presents significant challenges and limitations that must be addressed before widespread deployment in safety-critical environments. This

section critically examines these obstacles and outlines key future research directions.

A. Reliability and Safety Assurance

The paramount concern for autonomous vehicles is safety. The inherent characteristics of LLMs introduce new complexities.

- **Hallucinations and Factual Accuracy:** LLMs can generate factually incorrect or nonsensical information (hallucinations), which is unacceptable in AD where decisions must be precise and grounded in reality. Ensuring the reliability of LLM outputs, especially when used for planning or explanation, is a major hurdle [71].
- **Robustness to Adversarial Attacks and Deception:** LLMs are susceptible to adversarial prompts and jailbreaking attacks [35], which could compromise AD system safety if malicious input leads to unsafe behaviors. Research into LLM deceptive capabilities highlights these risks [33].
- **Guardrails and Alignment:** Developing robust guardrails and alignment techniques is crucial to ensure LLMs adhere to safety protocols and ethical guidelines [137]. Frameworks like NeMo Guardrails [4] offer toolkits for controllable and safe LLM applications, but their applicability to real-time, safety-critical AD scenarios needs further validation. The internal state of an LLM knowing when it's lying [138] offers a glimmer of hope for self-correction.
- **System-Level Reliability:** Integrating LLMs into complex AD architectures necessitates a holistic view of system-level reliability, considering the convergence of communication, computing, and control [139]. Cybersecurity and functional safety must be synergistically developed for Smart Electric Vehicles to mitigate risks from interconnected systems [72]. General reliability testing strategies from software engineering also become relevant [140]. For example, predicting the reliability of components, even in unrelated fields like blockchain peers [141], informs the need for robust reliability engineering.

B. Interpretability and Trust

The black-box nature of many LLMs hinders transparency, making it difficult to debug failures or ensure human trust.

- **Explainable AI (XAI) for AD:** While LLMs can generate explanations [81], the explanations themselves need to be accurate, relevant, and causally sound. Research on holistically aligned interpretability in end-to-end AD systems, like Hint-AD [82], is a step towards this. Crafting LLMs for enhanced interpretability, such as Concept Bottleneck LLMs, offers a path to inherently interpretable models [84].
- **Human Trust and Acceptance:** For autonomous vehicles to be widely adopted, users must trust their decisions. LLM-generated explanations can foster this trust, but the consistency and quality of these explanations are crucial [17].

C. Real-time Performance and Efficiency

LLMs are computationally intensive, posing challenges for real-time operation in autonomous vehicles.

- **Computational Demands:** Deploying large LLMs on vehicle hardware, which typically has limited computational and energy resources, is challenging. Significant research focuses on efficient architectures and inference for LLMs [25], [26]. Knowledge distillation (as discussed in Section III) is a key strategy for deploying LLM-derived intelligence on edge devices.
- **Latency:** AD systems require extremely low-latency responses. The sequential nature of LLM generation and potential for long inference times can be a bottleneck for critical decision-making. Techniques for real-time state monitoring [66] and efficient packet forwarding in vehicular ad hoc networks (VANETs) [142]–[144] are relevant to the underlying communication infrastructure required for prompt responses. Efficient task offloading and resource allocation in vehicular edge computing also address these latency constraints [145].

D. Data Requirements and Generalization

LLMs require vast and diverse datasets for training, and their generalization capabilities to novel AD scenarios are still under investigation.

- **Data Scarcity and Quality:** High-quality, annotated AD datasets are expensive to acquire. While LLMs can generate synthetic data [117], [118], ensuring the realism and diversity of this data, especially for long-tail events, remains critical [45], [146].
- **Sim2Real Gap:** Bridging the gap between simulated environments (where LLMs can be trained) and the complexities of the real world is a persistent challenge [69].
- **Multilingual and Low-Resource Scenarios:** While LLMs demonstrate strong multilingual capabilities [2], [8], performance in low-resource languages or culturally specific driving contexts can still lag [10].

E. Future Directions

Addressing the aforementioned challenges opens up numerous promising research avenues:

- **Robust Grounding of LLMs:** Developing mechanisms for LLMs to deeply understand and interact with the physical world, moving beyond textual descriptions to real-world sensory inputs and actions. This involves improved multi-modal fusion and representation learning [28]–[32].
- **Efficient and On-Device LLM Architectures:** Continued innovation in compact and efficient LLM architectures [25] and specialized hardware (e.g., edge AI accelerators) to enable real-time, low-power inference on autonomous vehicles.
- **Neuro-Symbolic Hybrid Systems with LLMs:** Further integrating LLMs with traditional symbolic AI for planning and knowledge representation, allowing LLMs to infer and utilize structured world models. This can leverage LLM reasoning as planning with world models [11] and improve self-consistency through probabilistic tokenization [147].
- **Continual Learning and Adaptation:** Empowering LLM-driven AD agents to continuously learn and adapt to new

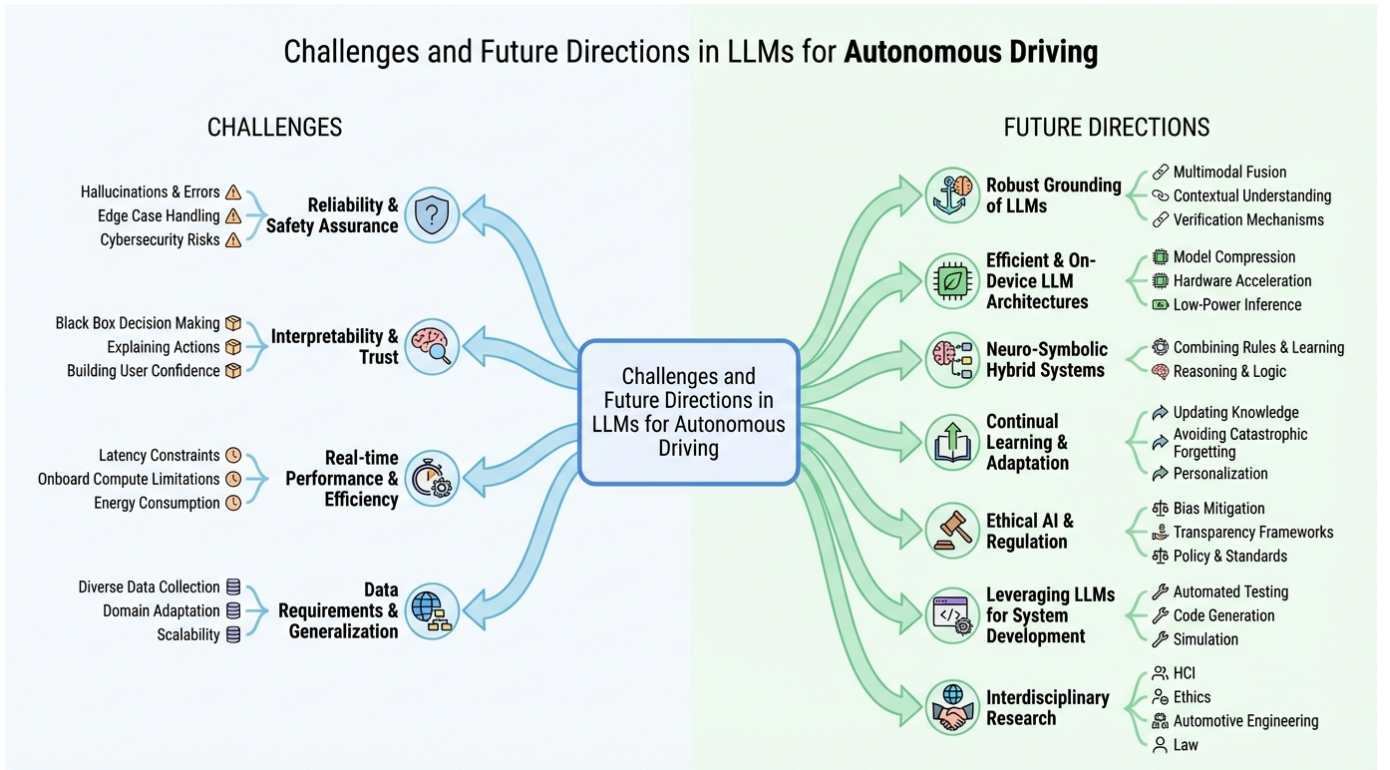


Fig. 4. Overview of key challenges and promising future directions for integrating Large Language Models into autonomous driving systems. Challenges include reliability, interpretability, real-time performance, and data requirements, while future directions focus on robust grounding, efficient architectures, hybrid systems, continual learning, ethical AI, system development leveraging, and interdisciplinary research.

driving scenarios and user preferences in the real world [103]. Exploring self-cognition in LLMs [36] could lead to more reflective and adaptive autonomous agents.

- **Ethical AI and Regulation:** Developing comprehensive ethical frameworks and regulatory guidelines specifically for LLM-enhanced autonomous systems, focusing on accountability, fairness, and transparency. This includes general AI safety summits [33].
- **Leveraging LLMs for System Development and Resource Management:** Expanding the role of LLMs in code generation and automation for AD systems [129]. Their analytical capabilities, similar to those for economic modeling [40]–[42] or supply chain optimization, could be used for fleet management, energy optimization, or predictive maintenance of autonomous vehicles. The underlying AI models can also contribute to data integrity through methods like image watermarking and forgery detection, which are increasingly relevant for synthetic AD data [148]–[150]. Advanced techniques for robust probabilistic distributional similarity can further aid in validating synthetic data and ensuring the consistency of data sets [151].
- **Interdisciplinary Research:** Fostering collaborations between NLP, robotics, control theory, human factors, and cognitive science to develop holistic solutions for LLM-driven autonomous driving. The general concept of building robust and adaptive frameworks, such as modular cross-domain adaptive templates, is crucial for scalable AI solutions [136].

Furthermore, while seemingly disparate, advances in fields like optics and microscopy [152]–[154] demonstrate the broad applicability of advanced computational techniques and inspire novel approaches to data processing. Even human behavioral modeling from unrelated domains like fatigue-aware ad creative selection [155] can offer insights into predicting human responses to AD systems.

VI. CONCLUSION

This survey has provided a comprehensive overview of the nascent yet rapidly evolving field at the intersection of Large Language Models and autonomous driving. We have established that the advanced natural language understanding, generative, and reasoning capabilities of LLMs offer a transformative potential to address complex challenges within AD systems, particularly in areas requiring high-level cognition, human-like interaction, and nuanced contextual understanding.

We explored the foundational aspects of both LLMs and autonomous driving, demonstrating how LLMs can serve as high-level planners, provide natural language interfaces for human-vehicle interaction, and act as multimodal reasoning engines by integrating diverse sensor inputs. Key applications discussed include enhancing scene understanding and semantic reasoning, generating human-centric explanations, assisting in complex planning, simulating human-like traffic behaviors, and supporting the development of AD software through code generation. Furthermore, we highlighted the potential for

knowledge distillation to bring LLM intelligence to resource-constrained autonomous platforms.

However, the path to widespread deployment of LLM-powered autonomous vehicles is fraught with significant challenges. Critical concerns include ensuring the reliability, safety, and interpretability of LLM decisions in safety-critical contexts, overcoming computational limitations for real-time performance, and addressing the intensive data requirements and generalization issues inherent in both LLMs and AD. Ethical considerations, including bias and the need for robust guardrails, also remain paramount.

Looking ahead, the future of LLMs in autonomous driving is bright, contingent upon continued interdisciplinary research. Key directions involve developing robust grounding mechanisms for LLMs in the physical world, engineering more efficient and on-device architectures, fostering advanced multimodal fusion, and exploring continuous learning paradigms. By synergistically combining the symbolic and cognitive strengths of LLMs with the robust perception and control of autonomous systems, we can pave the way for a new generation of intelligent vehicles that are safer, more capable, and seamlessly integrated into human society.

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